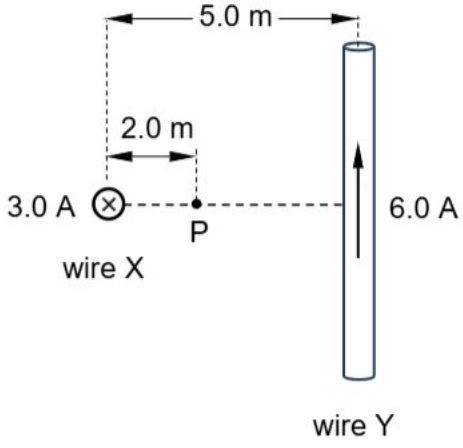
24 Two long, straight, current-carrying wires, X and Y are at a perpendicular distance 5.0 m apart. The current in X is 3.0 A and into the plane of the paper, while current in Y of 6.0 A is towards the top of the page.

Point P is on the line of shortest distance joining the two wires, at a distance 2.0 m away from wire X as shown.



What is the magnitude of the resultant magnetic flux density at point P?

A 3.5×10^{-12} T

B 5.0×10^{-7} T

C 6.3×10^{-7} T

D 7.0×10^{-7} T

Ans: **B**

Magnetic flux density is a vector.

$$B_x = \frac{\mu_0 I_x}{2\pi d_x} = \frac{(4\pi \times 10^{-7})(3.0)}{2\pi(2.0)} = 3.0 \times 10^{-7} \text{ T (downwards towards bottom of page)}$$

$$B_y = \frac{\mu_0 I_y}{2\pi d_y} = \frac{(4\pi \times 10^{-7})(6.0)}{2\pi(3.0)} = 4.0 \times 10^{-7} \text{ T (out of the page)}$$

$$B_{\text{resultant}} = \sqrt{B_x^{\,2} + B_y^{\,2}} = \sqrt{(3.0 \times 10^{-7})^2 + (4.0 \times 10^{-7})^2}$$
$$= 5.0 \times 10^{-7} \text{ T}$$

A: took epsilon instead.

C: swopped distances

D: took direction addition.